Free Decay of a Linearly Damped Oscillator: A Benchmark Note

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Abstract

A unit-mass, unit-stiffness oscillator with viscous damping coefficient 0.5 is released from rest conditions in the bench fixture. We record the decay envelope and the zero crossings of the displacement x with velocity y.

1 Equations of motion

With the state vector (x, y) the dynamics read

$$\dot{x} = y, \tag{1}$$

$$\dot{y} = -x - 0.5\,y. \tag{2}$$

2 Release conditions

The arm is clamped at a displacement of one and a half units and released from rest: the initial state is $x(0) = 1.5$ and $y(0) = 0$. The motion is underdamped, so the amplitude envelope decays geometrically while the sign of x alternates each half period.

3 Stability note

Both eigenvalues have negative real part, so every trajectory converges to the origin regardless of the release displacement.